Spin-Stratum Type-Rigidity: Orthogonality of the ADE Gate and Closure of the Chiral Lift

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Abstract

The generation-split value $\epsilon = \frac{1}{10}$ is often presented as a derived prediction of the spectral stratigraphy of the corpus. This note isolates its exact logical status. First, $\epsilon = \frac{1}{10}$ is a theorem modulo the ADE case-selection gate: once a case is selected whose class-complete spectrum carries the external level ratio $3:2$, the generation-deficit normal form forces $\epsilon = \frac{1}{10}$ and $\kappa = \frac{5}{12}$ algebraically, with no further input. Second, the first-principles status of ϵ is therefore exactly the first-principles status of the gate. We then characterise the obstruction to deriving the gate as an arithmetic orthogonality. The selection of the $\sqrt{5}$-locus that carries the $3:2$ ratio requires the nontrivial element of $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$, the action $\sqrt{5} \mapsto -\sqrt{5}$. The only field automorphism supplied by the foundations is the Born–Infeld parity $c \mapsto q - c$, which is the restriction of complex conjugation $\zeta_q \mapsto \zeta_q^{-1}$ and fixes $\sqrt{5}$. In the compositum $K_q = \mathbb{Q}(\zeta_q, \sqrt{5})$ the two actions lie in distinct direct factors of the Galois group, hence are arithmetically orthogonal. Consequently the gate cannot be discharged from the present axioms without an additional arithmetic principle at the spin stratum. A bounded character-table audit of the binary icosahedral group $2I \cong \mathrm{SL}(2,5)$ establishes the type-rigidity the gate needs: the operations available at the spin stratum of the present corpus construction all fix $\mathbb{Q}(\sqrt{5})$, so their generated group lies in the $\sqrt{5}$-fixing subgroup of $\mathrm{Gal}(K_q/\mathbb{Q})$, which by closure cannot manufacture the outer automorphism that realises $\sqrt{5} \mapsto -\sqrt{5}$. The no-go for the ADE gate is therefore unconditional in the present construction; the same rigidity closes the chiral orientation note, the Lorentzian chiral lift inducing no action on $\mathbb{Q}(\sqrt{5})$ and hence [H-orient] unconditionally within the present spin stratum [13]. Only the abstract type-rigidity of the full recursive tower remains open, and it is broader than either application requires.

Keywords: non-injective projection; ADE case selection; binary icosahedral group; golden ratio; cyclotomic field; Galois action; Born–Infeld parity; generation split; Lorentzian chiral lift; metaplectic lift; Weil representation type; epistemic stratification.

Contents

1 Introduction

The generation sector of the corpus produces a three-level deficit structure whose outer ratio matches the external ADE ratio $3 : 2$ of a selected case, and this matching fixes the split value $\epsilon = \frac{1}{10}$ and the common scale $\kappa = \frac{5}{12}$ [14, 12, 17]. The value $\epsilon = \frac{1}{10}$ has occasionally been described as derived. This note draws the line precisely, because the distinction between a posited value and a value forced once a case is selected is the whole content of the matter.

The clarification is in two parts. The first part is a trivial decomposition that nonetheless settles the status question: the algebra that takes the $3 : 2$ ratio to $\epsilon = \frac{1}{10}$ is general and case-independent, so all of the first-principles content lives in the upstream selection of the case (Section 2). The second part characterises why that upstream selection is not supplied by the present foundations. The selection of the $\sqrt{5}$-locus that carries the $3 : 2$ ratio is an arithmetic act: it requires the Galois action that separates the two two-dimensional irreducible characters of the binary icosahedral group $2I$. The foundations supply only the Born–Infeld parity, which is complex conjugation on the central character and fixes the real quadratic field $\mathbb{Q}(\sqrt{5})$. These two actions sit in distinct direct factors of the Galois group of the natural compositum, so the parity non-injectivity is not a partial version of the required Galois action; it is orthogonal to it (Sections 3–5).

The recursive non-injectivity theorem of ENI [5] guarantees that each stratum of a tower of genuinely distinct emergent levels carries its own non-injectivity, but it constrains existence, not arithmetic type. The base fibre is closed: its forced involution is the Born–Infeld parity [1]. The spin stratum is closed for the present corpus construction by a bounded character-table audit of $2I \cong \mathrm{SL}(2, 5)$, which shows that none of the operations available there realises the outer automorphism that moves $\sqrt{5}$ (Section 6, Appendix A). Whether every conceivable recursive stratum inherits the parity type, rather than introducing an action in the $\sqrt{5}$ factor, remains open at the abstract level, but this generalisation is broader than the ADE gate requires. The same spin-stratum type-rigidity has a second consequence, of opposite sign. The chiral orientation note closes [H-orient] in the inherited Heisenberg central-phase lift and isolates a single residual obstruction: a possible additional spin-Galois phase of the full Lorentzian chiral lift, orthogonal to ζ_q, in particular a $\sqrt{5}$ or ADE-spin factor [13]. Section 7 discharges that obstruction from the same Proposition 1: the Lorentzian lift induces no action on $\mathbb{Q}(\sqrt{5})$, the chiral weight σ_L being the Weyl conjugation type that lives in the parity factor and not in the $\sqrt{5}$ factor. For the gate the rigidity is fatal, the required $\sqrt{5}$ action being absent; for the chiral lift it is the closure, the feared $\sqrt{5}$ action being absent.

Section 8 records the epistemic stratification and the downstream consequence: observable masses route through the explicit operator E_Π and the Yukawa sector regardless of how the gate is resolved [10, 19].

2 The conditional split

We work with the generation-deficit normal form of the dynamical spectrum on the generation factor [14]. Let s_0 be the central eigenvalue, fixed by the projective involution J_Π, and let $s_\pm$ be the two outer eigenvalues, in the normalisation $s_0 = 1$.

Definition 1 (Exit deficit)**.** For an eigenvalue s with central value s_0, the exit deficit is $d_{\text{exit}}(s) := s_0 - s$. The normal form $s_0 = 1$, $s_\pm = \frac{1}{2} \pm \epsilon$ gives $d_{\text{exit}}(s_0) = 0$ and $d_{\text{exit}}(s_\pm) = \frac{1}{2} \mp \epsilon$.

Lemma 1 (Algebraic ratio lemma)**.** *If the outer exit deficits are in the ratio* $3:2$*, that is*

$$\frac{d_{\text{exit}}(s_-)}{d_{\text{exit}}(s_+)} = \frac{\frac{1}{2}+\epsilon}{\frac{1}{2}-\epsilon} = \frac{3}{2},$$

then $\epsilon = \frac{1}{10}$*. If in addition a single common scale* κ *satisfies* $\kappa\, d_{\text{exit}}(s_\pm) = |\lambda - 1|$ *on the two outer levels with* $\{|\lambda_1 - 1|, |\lambda_3 - 1|\} = \{\frac{1}{6}, \frac{1}{4}\}$*, then* $\kappa = \frac{5}{12}$*.*

Proof. From $2(\frac{1}{2}+\epsilon) = 3(\frac{1}{2}-\epsilon)$ one obtains $1+2\epsilon = \frac{3}{2}-3\epsilon$, hence $5\epsilon = \frac{1}{2}$ and $\epsilon = \frac{1}{10}$. The outer deficits are then $\{\frac{2}{5}, \frac{3}{5}\}$, and $\kappa = (\frac{1}{4})/(\frac{3}{5}) = (\frac{1}{6})/(\frac{2}{5}) = \frac{5}{12}$, consistent on both levels. □

The lemma is trivial and entirely case-independent: it uses only the ratio $3:2$, not any property of the binary icosahedral group or of the generating set. The status of $\epsilon = \frac{1}{10}$ is therefore the status of the hypothesis that supplies the ratio $3:2$.

Theorem 1 (Conditional ADE split)**.** *Assume the ADE case-selection gate selects the central-lift class over the* $\sqrt{5}$*-locus, that is the parity class over the* A_5 *five-cycle locus, whose class-complete Cayley spectrum has external ratio*

$$\frac{|\lambda_3 - 1|}{|\lambda_1 - 1|} = \frac{3}{2}$$

with $\{|\lambda_1 - 1|, |\lambda_3 - 1|\} = \{\frac{1}{6}, \frac{1}{4}\}$ *and central level* $\lambda_2 = 1$ *[6, 12]. Then the generation-deficit normal form forces, by Lemma 1,*

$$\frac{\frac{1}{2}+\epsilon}{\frac{1}{2}-\epsilon} = \frac{3}{2}, \qquad \epsilon = \frac{1}{10}, \qquad \kappa = \frac{5}{12}.$$

The split value is not an independent parameter once the ADE case is selected.

Remark 1 (Status)*.* Theorem 1 is conditional, not predictive. Its single nontrivial hypothesis is the gate. The first-principles status of $\epsilon = \frac{1}{10}$ is exactly the first-principles status of the gate, and nothing in the theorem bears on why the $\sqrt{5}$-locus is selected. The wording avoids committing to "order five" versus "order ten" for the generating class: the variable-valence exit analysis uses the order-five generating set [6], while the bounded-flux admissibility analysis uses the order-ten classes $10a \cup 10b$ [8], and the two are identified by the central lift, as made precise in Corollary 1.

3 The arithmetic organizing lemma

Two arithmetic data enter the construction from independent parts of the apparatus. The root of unity $\zeta_q = e^{2\pi i/q}$ is the central character of the fibre group $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, fixed by the Stone–von Neumann identification of the Weil representation V_ρ [1, 4]. The quadratic surd $\sqrt{5}$ is the character field of the two two-dimensional irreducible representations of the binary icosahedral group $2I$, where it appears through the golden ratio identity $\chi(10a)+\chi(10b) = \varphi - \bar{\varphi} = 1$ on the order-ten classes [8, 7]. These origins are independent: the first is a property of the fibre group on which the axioms act, the second is a representation-theoretic property of the spin-stratum subgroup.

Lemma 2 (Arithmetic orthogonality)**.** *Let* q *be prime with* $q \neq 5$*, and let* $K_q = \mathbb{Q}(\zeta_q, \sqrt{5})$*. Then* $\mathbb{Q}(\zeta_q)$ *and* $\mathbb{Q}(\sqrt{5})$ *are linearly disjoint over* $\mathbb{Q}$*, so*

$$\mathrm{Gal}(K_q/\mathbb{Q}) \;\cong\; \mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q}) \times \mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q}).$$

The Born–Infeld parity $c \mapsto q - c$ *is the restriction of complex conjugation; as an element of* $\mathrm{Gal}(K_q/\mathbb{Q})$ *it is*

$$\textit{BI parity} = (\zeta_q \mapsto \zeta_q^{-1},\ \sqrt{5} \mapsto \sqrt{5}).$$

The Galois action separating the two two-dimensional irreducible characters of $2I$ is

$$\textit{Galois-spin} = (\zeta_q \mapsto \zeta_q,\ \sqrt{5} \mapsto -\sqrt{5}).$$

These lie in distinct direct factors. In particular the Born–Infeld parity fixes $\mathbb{Q}(\sqrt{5})$ pointwise and is not the Galois-spin action.

Proof. The unique quadratic subfield of $\mathbb{Q}(\zeta_q)$ is $\mathbb{Q}(\sqrt{q^*})$ with $q^* = (-1)^{(q-1)/2}q$. For prime $q \neq 5$ one has $q^* \neq 5$ up to squares, so $\sqrt{5} \notin \mathbb{Q}(\zeta_q)$ and $\mathbb{Q}(\zeta_q) \cap \mathbb{Q}(\sqrt{5}) = \mathbb{Q}$. Linear disjointness gives the direct-product decomposition of the Galois group. The central character satisfies $e^{2\pi ic/q} \mapsto e^{2\pi i(q-c)/q} = e^{-2\pi ic/q} = \overline{e^{2\pi ic/q}}$ under $c \mapsto q - c$, so the Born–Infeld parity is complex conjugation on the ζ_q factor; since $\sqrt{5}$ is real, complex conjugation fixes it. The Galois-spin action is the nontrivial element of $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$ and fixes ζ_q. The two thus occupy distinct direct factors. □

Remark 2 (Why the disjunction is not accidental). The direct-factor separation reflects the independence of the two origins. The axioms act on the fibre $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ and on its cyclotomic factor $\mathbb{Q}(\zeta_q)$; they have no action on the factor in which $\sqrt{5}$ lives. The orthogonality is therefore structural, not numerical.

4 The central-lift tie-break

The order-ten elements of $2I$ are the order-five elements multiplied by the central element $-I$, since for an order-five g one has $(-Ig)^5 = -I$ and $(-Ig)^{10} = I$. The Weil lift of $-I$ is the central reflection $R_b = W(-I) = \mathrm{FT}^2$, scalar on the chiral carrier and commuting with γ_5 [18].

Corollary 1 (Order-five versus order-ten is below parity resolution)**.** *Under the identification $R_b = W(-I)$ of the central lift with the Born–Infeld parity, the projective quotient that the construction realises identifies the order-five and order-ten classes rather than distinguishing them. The order-five versus order-ten alternative is not a selectable degree of freedom of the framework; it is a central-lift ambiguity that the parity erases.*

Proof. The pair of classes is exchanged by $\times(-I)$, whose realisation is R_b. By Lemma 2 the parity acts only on the ζ_q factor and fixes $\sqrt{5}$, so it permutes the classes by the central lift while leaving the Galois data invariant. Quotienting by the central lift identifies the two classes; it does not order them. Hence the framework supplies no selector between order five and order ten, for the same arithmetic reason that it supplies no Galois-spin action: it provides $\times(-I)$, not $\sqrt{5} \mapsto -\sqrt{5}$. □

The tie-break and the negative result of Section 5 are thus a single fact. The parity fixes $\mathbb{Q}(\sqrt{5})$ and permutes only by $\times(-I)$. It therefore neither separates order five from order ten, nor separates the two Galois-conjugate characters $5a$ and $5b$.

5 The negative theorem

Theorem 2 (No Galois-spin selection for the ADE gate in the present corpus construction)**.** *For the spin stratum of the present corpus construction (Proposition 1), every field automorphism forced by the foundations A1–A4, the recursive non-injectivity of ENI Corollary 6 [5] and the A4 locking lies in the factor* $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$ *of* $\mathrm{Gal}(K_q/\mathbb{Q})$ *and fixes* $\mathbb{Q}(\sqrt{5})$ *pointwise. Consequently the Galois-spin selection, the nontrivial element of* $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$*, is not supplied by these operations. The* $\sqrt{5}$*-locus required by the ADE* $3:2$ *dictionary therefore cannot be selected, in the present construction, without an additional arithmetic principle at the spin stratum.*

Proof. By Proposition 1 the operations available at the spin stratum each fix $\mathbb{Q}(\sqrt{5})$, so by Lemma 2 their generated group lies in $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$. The Galois-spin action is the nontrivial element of the complementary factor $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$, which is not in the image of these operations. Selecting the $\sqrt{5}$-locus over the neutral order-four locus distinguishes the two two-dimensional characters by their $\sqrt{5}$ content, hence requires that complementary factor, which the present construction does not provide. □

Remark 3 (Scope). Theorem 2 is unconditional for the ADE gate in the present construction, resting on the established Proposition 1. The abstract generalisation to every conceivable recursive stratum is Lemma 3, which remains open and is broader than the gate requires.

Remark 4 (Selection against, not merely absence of selection). The neutral generators derived from Born–Infeld parity and admissibility are those with vanishing spin-$\frac{1}{2}$ character, the order-four half-turns; the exact penalised spectral-capacity computation confirms that the order-five representative of the $\sqrt{5}$-locus is dominated and not selected for $\alpha > 0$ [8, 9]. The negative result is therefore an orthogonality between the supplied parity and the required Galois action, not a mere gap in the derivation.

6 Type-rigidity along the tower

Proposition 1 (Spin-stratum type-rigidity for the corpus construction)**.** *Let $\mathcal{A}_{\text{spin}}$ be the group generated by the operations available at the spin stratum of the present corpus construction: inner conjugations in $2I$, the central element $-I$, the central reflection $R_b = W(-I)$, the projective involution J_Π, the tensor functors of $\mathrm{Rep}(2I)$, the Schur projection, and the A4 locking. Each generator fixes $\mathbb{Q}(\sqrt{5})$ pointwise, so $\langle \mathcal{A}_{\text{spin}} \rangle$ is contained in the $\sqrt{5}$-fixing subgroup of $\mathrm{Gal}(K_q/\mathbb{Q})$. In particular it does not contain the outer automorphism of $2I \cong \mathrm{SL}(2,5)$, the unique automorphism that realises $\sqrt{5} \mapsto -\sqrt{5}$ and exchanges the two two-dimensional irreducible characters.*

Proof. The action of each generator on $\mathrm{Irr}(2I)$ is determined by a character-table audit of $2I \cong \mathrm{SL}(2,5)$, carried out from the group in Appendix A and validated by the orthogonality relations. Inner conjugations act trivially on $\mathrm{Irr}(2I)$, since characters are class functions. The central element $-I$ and its Weil lift $R_b = W(-I)$ act on each irreducible representation by the scalar $\chi(-I)/\chi(1) = \pm 1$, a spinorial grading that permutes nothing. The projective involution J_Π is complex conjugation on the central character; since every character of $2I$ is real-valued, it fixes every irreducible representation. The tensor functors have integer fusion coefficients, and the Schur projection is defined through the J_Π-parity and symplectic data, so both fix $\mathbb{Q}(\sqrt{5})$. The A4 locking supplies an orientation, but it is the spontaneous $V-A$ sign $u(\pm s_*) = \mp u(s_*)$ under the parity $s \mapsto -s$ [17]; this is the J_Π-odd chiral $\mathbb{Z}/2$, in the parity factor, not the Galois action $\sqrt{5} \mapsto -\sqrt{5}$, and the magnitude $|u|$ is itself dictionary-bound through $\mathcal{N}_A$. Each generator therefore fixes $\mathbb{Q}(\sqrt{5})$. Since the $\sqrt{5}$-fixing automorphisms form a subgroup, closed under composition and inverse, no iteration, central lift, projection or A4 locking built from these generators can manufacture the outer automorphism, which moves $\sqrt{5}$. □

Lemma 3 (Full recursive type-rigidity, open)**.** *Every non-injective projection forced by A1–A4 and ENI Corollary 6, including the recursive non-injectivities at every stratum above the base fibre, acts on the cyclotomic factor $\mathbb{Q}(\zeta_q)$ and fixes the real quadratic factor $\mathbb{Q}(\sqrt{5})$.*

Three regimes of this statement have different status, and conflating them is exactly the error to avoid.

For the base fibre Π_1 the statement is established. The forced fibre involution is the Born–Infeld parity, the minimal fibre condition $\Pi^{-1}(y) \supseteq$

$\{\chi, -\chi\}$ closed by A3 and the Stone–von Neumann identification of V_ρ [1, 4]. By Lemma 2 this involution lies in $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$ and fixes $\sqrt{5}$.

For the spin stratum the statement is established for the present corpus construction by Proposition 1: the operations actually available there generate a $\sqrt{5}$-fixing group. This is the only stratum the $\sqrt{5}$-locus of the ADE gate requires, so the no-go of Theorem 2 is unconditional in the present construction.

For the abstract claim over every conceivable recursive stratum the statement is open. ENI Corollary 6 forces $S_{\Pi_i} > 0$ at each stratum of a tower of genuinely distinct emergent levels, but it constrains existence of the non-injectivity, not its arithmetic type [5, 11]. Nothing in the corollary forbids some other construction from realising a stratum non-injectivity by an automorphism in the $\sqrt{5}$ factor.

Remark 5 (Proof direction for the full-tower version, not concluded). The entropy monotonicity $S_{\Pi_2 \circ \Pi_1} \geq S_{\Pi_1}$ of ENI Corollary 6 constrains the composition of strata. If the only automorphism available to realise a fibre non-injectivity is the parity, a candidate argument is that the composition cannot create an arithmetic factor absent from the individual stages, so that the recursive type is inherited from the base. This is a type-inheritance claim, distinct from entropy preservation, and the step from one to the other is not filled. Lemma 3 is therefore stated as structural and open; it is broader than the ADE gate requires, and Theorem 2 does not depend on it.

7 The chiral-lift corollary

The spin-stratum type-rigidity of Proposition 1 was used in Section 5 to obstruct the ADE gate. The same proposition discharges the residual obstruction of the chiral orientation note, where it acts with the opposite sign. There [H-orient] is closed in the inherited Heisenberg central-phase lift, the chiral phase being the linear cyclotomic character $\omega_\chi(e) = \zeta_q^{\Delta A_c(e)}$, and the single caveat left is a possible additional spin-Galois phase of the full Lorentzian chiral lift, orthogonal to ζ_q [13]. We show the caveat does not arise.

The object is the metaplectic lift of the ordered cascade step. The two cascade flows are the Weil images of the unipotent one-parameter subgroups of $\mathrm{SL}(2, \mathbb{R})$, with product $S_X(t)S_Y(s) = W\left(\begin{smallmatrix} 1+ts & t \\ s & 1 \end{smallmatrix}\right)$; a Baker–Campbell–Hausdorff reduction identifies the per-step J_3 component with the oriented symplectic area swept per step, the Carnot-degree-two central increment of Heis_3 [15]. The Lorentzian metric forces the complexification $\mathfrak{mp}(2, \mathbb{R})_\mathbb{C} \simeq$

$\mathfrak{sl}_2(\mathbb{C})$ and the chiral split $S_L \oplus S_R$ [2, 16], and the chiral weight σ_L classes S_L versus S_R by the Weyl representation type $\mathbf{2}$ versus $\bar{\mathbf{2}}$ [13].

Lemma 4 (The Lorentzian boost is Galois-internal)**.** *The action induced on K_q by the Lorentzian chiral lift extends $\mathcal{A}_{\mathrm{spin}}$ only by the central character and the analytic complexification, both of which fix $\mathbb{Q}(\sqrt{5})$. The central character contributes the cyclotomic phase ζ_q, whose only field automorphism is complex conjugation $\zeta_q \mapsto \zeta_q^{-1}$ [3]. The metaplectic cocycle is absent in the inherited model, and should it appear it is cyclotomic of Weil/Maslov type and does not supply $\sqrt{5} \mapsto -\sqrt{5}$ [13]. The Lorentzian boost is the analytic complexification unit i on the integral structure constants of $\mathfrak{sl}_2(\mathbb{C})$, inducing no automorphism of K_q outside the cyclotomic factor. Hence the boost lies in the $\sqrt{5}$-fixing subgroup of $\mathrm{Gal}(K_q/\mathbb{Q})$: it is Galois-internal to the spin stratum.*

Lemma 5 (Weyl conjugation versus Galois conjugation)**.** *The chiral weight σ_L classes the Weyl branches S_L versus S_R by complex-conjugation type, $\mathbf{2}$ versus $\bar{\mathbf{2}}$ [13], whereas the Galois action $\sqrt{5} \mapsto -\sqrt{5}$ classes the two two-dimensional irreducible characters 2 versus $2'$ of $2I$, the swap realised by σ alone (Appendix A). These distinctions are not the same. Under the restriction $2I \subset \mathrm{SU}(2) \subset \mathrm{SL}(2, \mathbb{C})$ the representations of $\mathrm{SU}(2)$ are pseudoreal, so S_L and S_R restrict to the same two-dimensional carrier, while $2'$ is the Galois conjugate of that carrier and not the right Weyl branch. The spinorial grading $\chi(-I)/\chi(1) = -1$ that σ_L reads is common to both two-dimensional irreducibles (Proposition 1), and their $\sqrt{5}$ content sits only on the order-five and order-ten classes (Corollary 1), which the Weyl-conjugation label does not address. Hence σ_L is orthogonal to the outer automorphism that distinguishes 2 from $2'$.*

Corollary 2 (Spin-Galois type-rigidity of the Lorentzian chiral lift)**.** *Let q be prime with $q \neq 5$. Every field automorphism induced on K_q by the Lorentzian chiral lift of the ordered cascade step acts trivially on $\mathbb{Q}(\sqrt{5})$; the nontrivial action, when present, is cyclotomic. In particular σ_L lies in the parity factor $\mathrm{Gal}(\mathbb{Q}(\zeta_q)/\mathbb{Q})$ and does not probe the Galois-spin factor $\mathrm{Gal}(\mathbb{Q}(\sqrt{5})/\mathbb{Q})$.*

Proof. By Proposition 1 the spin-stratum operations fix $\mathbb{Q}(\sqrt{5})$, and by Lemma 4 so does the Lorentzian boost; their generated group lies in the $\sqrt{5}$-fixing subgroup of $\mathrm{Gal}(K_q/\mathbb{Q})$, closed under composition and inverse. The only generator that could move $\sqrt{5}$ is the outer automorphism of $2I$, which by Lemma 5 is not coupled by σ_L and is supplied by no operation of the lift. Hence the induced action on K_q fixes $\mathbb{Q}(\sqrt{5})$, and any nontrivial part is cyclotomic. $\square$

Corollary 3 (Unconditional orientation within the present spin stratum)**.** *Within the Lorentzian chiral lift generated by the present spin stratum,*

$$\omega_\chi(e) = \zeta_q^{\Delta A_c(e)} \implies \rho_\chi = 1 \implies \text{[H-orient]} \implies N_A = \theta_{\max} \neq 0.$$

Proof. By Corollary 2 the chiral phase carries no Galois-spin factor orthogonal to ζ_q, so the residual obstruction left open by the chiral orientation note [13] does not arise. After the central-character convention and the outgoing branch of [13] are fixed, a cyclotomic reindexing does not define a new spin-Galois phase: it is absorbed into the choice of the Heisenberg central block and its Born–Infeld conjugate. Thus the relevant chiral landing phase is the same linear character $\omega_\chi(e) = \zeta_q^{\Delta A_c(e)}$ used in the inherited-lift calculation, rather than an independent average over cyclotomic embeddings. The implications are then those of the inherited lift, now with no orthogonal factor available to set $\rho_\chi = 0$. □

Remark 6 (Boundary with the Schur transport)*.* Corollary 3 closes the frontier amplitude datum $N_A \neq 0$. It does not establish $u \neq 0$, which lives in the projected Dirac residue through the Schur complement, as the J_Π-odd asymmetry of the eliminated block and not in any frontier average [19]. The statement closed here is $N_A \neq 0$, not $\Pi_S\, D\, (\partial_s P|_0)\, D\, \Pi_S^*\big|_{\text{gen},\, J_\Pi\text{-odd}} \neq 0$; the promotion of the signal to the spectral split $u \neq 0$ remains the Schur transport on the J_Π-odd generation sector [19, 12].

8 Status and downstream consequence

The epistemic stratification of the nine statements is as follows.

Statement	Content	Status
Lemma 1	ratio $3:2 \Rightarrow \epsilon = \frac{1}{10}$	proved, trivial, case-independent
Theorem 1	gate $\Rightarrow$ ratio $3:2 \Rightarrow \epsilon$	conditional / open hypothesis
Lemma 2	parity $\perp$ Galois-spin in $\mathrm{Gal}(K_q/\mathbb{Q})$	proved
Corollary 1	order five = order ten mod central lift	proved, modulo $R_b = W(-I)$
Theorem 2	no Galois-spin for the ADE gate (present construction)	proved
Proposition 1	spin-stratum type-rigidity (corpus construction)	proved, character-table audit
Corollary 2	chiral lift fixes $\mathbb{Q}(\sqrt{5})$; $\sigma_L \perp$ Galois-spin	proved
Corollary 3	[H-orient] closed, $N_A \neq 0$ (present stratum)	proved
Lemma 3	full recursive (all-tower) type-rigidity	structural / open

The net statement is that $\epsilon = \frac{1}{10}$ is a theorem modulo the ADE case-selection gate, and that for the present corpus construction the gate is a no-go: the spin-stratum operations generate a $\sqrt{5}$-fixing subgroup $\langle \mathcal{A}_{\mathrm{spin}} \rangle \subset \mathrm{Gal}(K_q/\mathbb{Q})_{\sqrt{5}}$, closed under composition, so no iteration, central lift, Schur projection or A4 locking can manufacture the outer automorphism that the $\sqrt{5}$-locus requires. The obstruction is therefore not an unresolved computation but a closure argument: for the present corpus construction, the gate requires an additional arithmetic principle not supplied by A1–A4, ENI, or A4 locking. The same closure argument, applied to the Lorentzian chiral lift, discharges the residual obstruction of the chiral orientation note: the lift induces no action on $\mathbb{Q}(\sqrt{5})$, so [H-orient] is closed unconditionally within the present spin stratum, the gate no-go and the chiral closure being opposite cuts of Proposition 1 [13].

The downstream consequence bounds the relevance of the gate. The gate fixes a ratio of levels and hence the structural input ϵ; it does not fix observable masses. The charged-lepton and quark spectra and the mixing matrices require the explicit construction of the operator E_{Π} and the projected Yukawa sector, with the mass hierarchy carried by the cascade amplification and its exponent [10, 19, 12]. Even a derivation of the gate would not by itself produce a mass. The gate is therefore a finite, characterisable structural sub-problem, now settled negatively for the present construction, while the quantitative effort proceeds on E_{Π}, the explicit Lorentzian block, and the

Yukawa sector.

A Reproducible audit of the spin-stratum operations

The audit is a finite, bias-independent computation, reproducible from the group with no hard-coded character data. The protocol is as follows.

1. Construct $2I \cong \mathrm{SL}(2,5)$ explicitly as the 120 matrices over $\mathbb{F}_5$ of determinant one.

2. Compute the conjugacy classes from the group: nine classes, of orders $1, 2, 3, 4, 5, 5, 6, 10, 10$ and sizes $1, 1, 20, 30, 12, 12, 20, 12, 12$.

3. Reconstruct the character table from the class algebra by simultaneous diagonalisation of the class-sum matrices A_i, with $(A_i)_{jk} = a_{ikj}$ the class-multiplication constants, recovering the dimensions $1, 2, 2', 3, 3', 4, 4', 5, 6$.

4. Validate by the row orthogonality relations, which hold to machine precision; the reconstructed table agrees with the published character table of $2I$ [8].

5. Apply the Galois action $\sqrt{5} \mapsto -\sqrt{5}$ to the table: it induces the permutation $(2\,2')(3\,3')$ of $\mathrm{Irr}(2I)$, fixing $1, 4, 4', 5, 6$.

6. Compute the action of each spin-stratum operation: inner conjugations, complex conjugation J_Π, the central element $-I$ and $R_b = W(-I)$ all induce the identity permutation of $\mathrm{Irr}(2I)$; none realises $2 \leftrightarrow 2'$.

The two two-dimensional irreducible characters on the $\sqrt{5}$-locus are Galois-conjugate, with $\varphi = (1+\sqrt{5})/2$ and $\bar{\varphi} = \varphi - 1 = (\sqrt{5}-1)/2$:

	$5a$	$5b$	$10a$	$10b$
χ_2	$\bar{\varphi}$	$-\varphi$	φ	$-\bar{\varphi}$
$\chi_{2'}$	$-\varphi$	$\bar{\varphi}$	$-\bar{\varphi}$	φ

so $\chi_{2'} = \sigma(\chi_2)$ with $\sigma : \sqrt{5} \mapsto -\sqrt{5}$, and the swap $2 \leftrightarrow 2'$ is realised by σ alone. Since every character of $2I$ is real-valued, complex conjugation J_Π fixes each irreducible representation; the central element $-I$ and $R_b = W(-I)$ act by the scalar $\chi(-I)/\chi(1) = \pm 1$ without permuting; and inner conjugations are trivial on class functions. No spin-stratum operation moves $\mathbb{Q}(\sqrt{5})$, and the $\sqrt{5}$-fixing automorphisms form a subgroup, so no composition of them realises σ. This establishes Proposition 1. A companion script performing the construction, reconstruction, orthogonality validation and permutation tests is deposited with the note.

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